

Numpy+Pandas+Matplotlib

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# =====
# STUDENT DATA
# =====

data = {
    "Student": ["Asha", "Ravi", "Priya", "Kiran", "Sneha"],
    "Math": [85, 78, 92, 65, 88],
    "Science": [90, 82, 89, 70, 85],
    "English": [80, 75, 95, 60, 90]
}

# =====
# PANDAS DATAFRAME
# =====

df = pd.DataFrame(data)

# =====
# NUMPY CALCULATIONS
# =====

marks = df[["Math", "Science", "English"]].values

df["Total"] = np.sum(marks, axis=1)
df["Average"] = np.mean(marks, axis=1)

# Grade Assignment
grades = []

for avg in df["Average"]:
    if avg >= 90:
        grades.append("A")
    elif avg >= 80:
        grades.append("B")
    elif avg >= 70:
        grades.append("C")
    else:
        grades.append("D")

df["Grade"] = grades

# =====
# DISPLAY RESULTS
# =====

print("\nStudent Performance Report\n")
```

```
print(df)

print("\nClass Statistics")
print("Highest Average:", np.max(df["Average"]))
print("Lowest Average :", np.min(df["Average"]))
print("Class Average  :", np.mean(df["Average"]))

# =====
# MATPLOTLIB VISUALIZATION
# =====

plt.figure(figsize=(8,5))

plt.bar(df["Student"], df["Average"])

plt.title("Student Average Marks")
plt.xlabel("Students")
plt.ylabel("Average Marks")

plt.grid(True)

plt.show()
```

Student Performance Report

	Student	Math	Science	English	Total	Average	Grade
0	Asha	85	90	80	255	85.000000	B
1	Ravi	78	82	75	235	78.333333	C
2	Priya	92	89	95	276	92.000000	A
3	Kiran	65	70	60	195	65.000000	D
4	Sneha	88	85	90	263	87.666667	B

Class Statistics

Highest Average: 92.0

Lowest Average : 65.0

Class Average : 81.6

